CLINICAL EXAMINATION · PEDIATRIC MEDICINE SUMMARY

Pediatric Clinical *Examination*

General Examination (Appearance, Growth, Vital Signs & Blood Pressure, Head-to-Toe Survey) and Systemic Examination: Neurological (Pediatric GCS, Cranial Nerves, Reflexes), Cardiovascular (Murmurs), Respiratory (Signs of Distress) and Abdominal (Liver, Spleen & Ascites).

Lecture Summary · **Pediatric Physical & Clinical Examination**

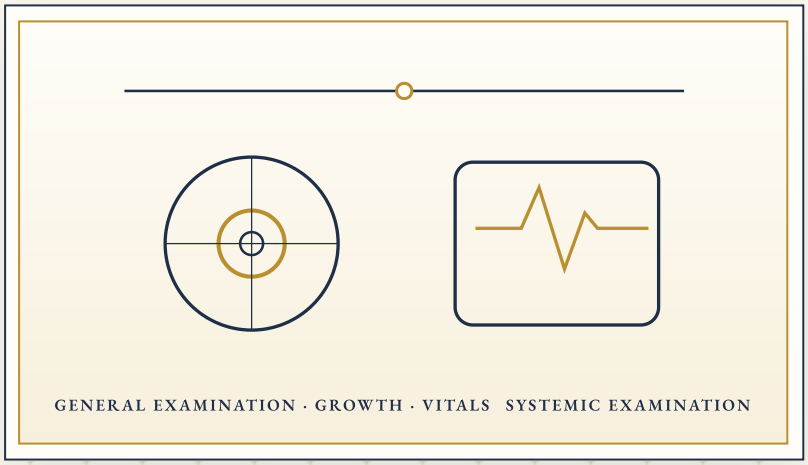


PLATE I — *Observe before you touch: general examination (left) and systemic examination of the nervous system, heart, chest and abdomen (right).*

Duas Before Studying

ابدأ مذاكرتك بهذه الأدعية المباركة لتيسير الفهم والحفظ والعلم النافع:

دعاء الفهم

اللَّهُمَّ إِنِّي أَسْأَلُكَ فَهْمَ النَّبِيِّينَ، وَحِفْظَ الْمُرْسَلِينَ، وَالْمَلَائِكَةِ الْمُقَرَّبِينَ

دعاء قبل المذاكرة

اللَّهُمَّ افْتَحْ عَلَيَّ فُتُوحَ الْعَارِفِينَ بِحِكْمَتِكَ، وَيَسِّرْ لِي أَمْرِي، وَاشْرَحْ لِي صَدْرِي بِمَعْرِفَتِكَ

دعاء العلم النافع

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا، وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا

دعاء سيدنا موسى عليه السلام

رَبِّ اشْرَحْ لِي صَدْرِي، وَيَسِّرْ لِي أَمْرِي، وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي

نية المذاكرة

نَوَيْتُ بِمُذَاكَرَتِي هَذِهِ خِدْمَةَ مَرْضَى الْمُسْلِمِينَ وَنَفْعَ أُمَّةِ نَبِيِّكَ مُحَمَّدٍ ﷺ

دعاء تيسير العلم

اللَّهُمَّ لَا سَهْلَ إِلَّا مَا جَعَلْتَهُ سَهْلًا، وَأَنْتَ تَجْعَلُ الْحَزْنَ إِذَا شِئْتَ سَهْلًا

إرشاد للمذاكرة (Pediatric Clinical Examination): يغطي هذا الملخص محاضرة (الفحص الإكلينيكي للأطفال) طبقاً لنص السلايدات، من غير أي إضافات خارجية.

1. Approach: Examine the Child, Not the Protocol

OBSERVE FIRST

- General appearance, color, affect and respiratory effort are judged **before any touch**.
- Count the **respiratory rate while the child is calm**.
- Watch how the child plays and interacts.

POSITION OF COMFORT

- Infants and toddlers can be examined **on the parent's lap**.
- Caregiver and child must be comfortable before you begin.
- Use distraction, toys and a **warm stethoscope**.

ADAPT THE ORDER

- Sequence may be modified by **age and cooperation**.
- **Quiet tasks first:** heart and lungs.
- Leave **ears, throat** and uncomfortable areas **to the end**.

طريقة فحص الطفل:

- **بُص قبل ما تلمس:** الشكل العام واللون والمجهود في التنفس بيتقيّموا من غير لمس، وعدّ التنفس والطفل هادي.
- **الرضيع ممكن يتفحص على رجل أمه،** واستخدم لعب، والسماعة تبقى **دافية.**
- **الحاجات الهادية الأول** (القلب والرئة)، **والودن والزور والحاجات اللي بتضايقه في الآخر.**

2. General Examination, Growth & Head-to-Toe Survey

1. General Examination: Six Steps

STEP	WHAT TO CHECK
1. Appearance	Alertness, color, distress · Hydration, nutrition · Interaction with parent
2. Growth	Weight, length or height · Head circumference in infants · Plot on charts
3. Vital signs	Heart rate, respiratory rate · Temperature, SpO_2 · **Blood pressure from 3 years**
4. Skin & nodes	Pallor, jaundice, rash · Edema: face and legs · Lymph node chains
5. Head & neck	Fontanelle, head shape · Eyes, ears, throat · Neck and thyroid
6. Hands & limbs	Clubbing, nails · Perfusion, capillary refill · Spine, hips, gait

Nephrology focus: periorbital and pedal edema, blood pressure, growth faltering, pallor, signs of rickets. Include the exam of the abdomen for ascites and masses.

2. Growth: Measure Accurately, Plot Every Time

— **Weight:** minimal clothing, calibrated scale.
— **Under 2 years: recumbent length** on a length board.
— **From 2 years: standing height** on a stadiometer.
— **Head circumference** (infants): widest occipito-frontal circumference.
— **BMI-for-age** from 2 years.
— **Correct age is critical**; plot on the chart for age and sex.

AGE	CHART
< 2 years	**WHO** Child Growth Standards (all feeding types)
2 – 20 years	**CDC** growth reference charts

WHO cut-offs: **2nd** and **98th** percentiles (short stature, low or high weight-for-length).

3. Head-to-Toe Survey: What to Look For

REGION	LOOK FOR
1. Head	Head circumference, shape, anterior fontanelle (infants)
2. Ears & eyes	Red reflex, tympanic membranes, periorbital edema
3. Mouth, throat & neck	Tonsils, lymph nodes, thyroid
4. Chest & heart	Precordium, work of breathing, pulses
5. Skin, hands & nails	Pallor, rashes, clubbing, capillary refill
6. Abdomen & hips	Organomegaly, masses, ascites, genitalia, hip stability
7. Limbs & spine	Edema, bowing, gait, spinal alignment

4. Step-by-Step: General Examination Technique

— **Recumbent length:** two examiners, head against the board, legs gently extended; record to **0.1 cm**.
— **Head circumference:** tape over the most prominent part of forehead and occiput; take the **largest of 3**.
— **Blood pressure:** rested, seated, right arm at heart level, correct cuff size.
— **Skin & lymph nodes:** inspect pallor and edema; palpate cervical, axillary and inguinal chains.

الفحص العام والنمو:

• **الفحص العام 6 خطوات:** الشكل ← النمو ← العلامات الحيوية (والضغط من 3 سنين) ← الجلد والغدد ← الراس والرقبة ← الإيدين والأطراف.
• **تحت سنتين:** الطول نايم وWHO charts. **من سنتين لـ 20 سنة:** الطول واقف وCDC charts، والـ BMI من سنتين.
• **محيط الراس:** أكبر قراءة من 3. **الطول نايم:** محتاج اتنين، ويتسجل لأقرب 0.1 سم.

3. Vital Signs & Blood Pressure

1. Vital Signs Vary with Age (PALS 2015)

AGE GROUP	HR AWAKE	HR ASLEEP	RR /MIN	SYSTOLIC BP
Infant (1 mo – 1 y)	100 – 190	90 – 160	30 – 53	72 – 104
Toddler (1 – 2 y)	98 – 140	80 – 120	22 – 37	86 – 106
Preschool (3 – 5 y)	80 – 120	65 – 100	20 – 28	89 – 112
School age (6 – 11 y)	75 – 118	58 – 90	18 – 25	97 – 115*
Adolescent (12 – 15 y)	60 – 100	50 – 90	12 – 20	110 – 131

**Systolic BP range shown for ages 6 to 9 years. HR in bpm, BP in mmHg.*

2. Blood Pressure in Children: AAP 2017

WHO & WHEN

— Measure BP **annually from 3 years** of age.

— **"Elevated BP"** replaces "prehypertension".

HOW

— **Right arm**, correct cuff size (bladder width ≈ **40%** of arm circumference).

— Oscillometric devices for screening; **confirm elevated readings by auscultation**.

CONFIRM

— **Ambulatory BP monitoring (ABPM)** confirms hypertension and suspected white-coat hypertension.

— New normative tables based on normal-weight children.

العلامات الحيوية والضغط:

• النبض والتنفس **يقلّوا كل ما السن يكبر**، والضغط بيعلى. مثلاً الرضيع تنفسه 30–53، والمراهق 12–20.

• **الضغط (AAP 2017)**: بيتقاس كل سنة من سن 3 سنين، على **الدراع اليمين**، وعرض الـ cuff حوالي 40% من محيط الدراع. القراءة العالية بالجهاز بتتأكد بالسماعة، والتشخيص النهائي بالـ ABPM. ومصطلح "Elevated BP" بدل "prehypertension".

4. Neurological Examination

1. Neurological Exam: A Mental Template (6 Steps)

— **1. History first:** family history, pregnancy & delivery, developmental milestones.

— **2. Level of consciousness:** alert, lethargic, obtunded, stuporous; pediatric GCS.

— **3. Cranial nerves:** through play (toys, light, bell, smile, tongue).

— **4. Motor system:** posture, bulk, tone, power, involuntary movements.

— **5. Reflexes:** deep tendon and, in infants, primitive reflexes.

— **6. Sensation, coordination, gait:** as cooperation allows.

TOOLS TO BRING

— Bell and bright toy · Ophthalmoscope

— Reflex hammer · Cotton-tipped applicator

— Measuring tape

> Examine the **skin** carefully: nervous system and skin share **ectodermal origin**.

2. Consciousness & the Pediatric Glasgow Coma Scale

SCORE	EYE	VERBAL (INFANT)	VERBAL (CHILD)	MOTOR
6	—	—	—	Obeys (child) / moves spontaneously (infant)
5	—	Coos, babbles	Oriented	Localizes pain
4	Spontaneous	Irritable cry	Confused	Withdraws
3	To voice	Cries to pain	Inappropriate words	Abnormal flexion
2	To pain	Moans to pain	Incomprehensible sounds	Abnormal extension
1	None	None	None	None

Levels: Alert · **Lethargy** = drowsy but responds · **Obtundation** = slow, confused · **Stupor** = wakes to pain only.

Total **3 – 15**. ≤ **12** → severe head injury; ≤ **8** → may need airway protection & ICP monitoring. Intubated or preverbal → **motor response matters most**.

3. Cranial Nerves Through Play

NERVE	CHILD-FRIENDLY TEST
II optic	Fixation and following, visual fields, red reflex, fundi
III, IV, VI	Follow a light or toy through the **H pattern** (six cardinal positions); pupils, ptosis
V trigeminal	Facial sensation, jaw clench; **rooting and sucking** in infants
VII facial	Symmetry of smile and cry, forehead wrinkling, eye closure
VIII	Startle to bell or clap; turns to voice
IX, X	Palate rise, gag, voice, swallowing
XI	Shoulder shrug, head turning against resistance
XII	Tongue in midline, fasciculations

4. Motor System, Reflexes & Primitive (Newborn) Reflexes

MOTOR & TONE

— **Infant:** posture, spontaneous movements, **head lag on pull-to-sit**.
— **Older child:** bulk, tone, **power 0 to 5**, gait and Gowers.

REFLEXES

— Deep tendon reflexes graded **0 to 4+**.
— Plantar **extensor (Babinski) is normal in young infants**.
— **Asymmetry matters more than the grade.**

NEWBORN REFLEX	HOW IT LOOKS
Babinski	Stroke the sole → big toe bends back and the other toes fan out
Rooting	Stroke the mouth → baby turns and opens the mouth (helps feeding)
Moro ("startle")	Sudden loud noise or movement → baby extends the arms, palms up
Sucking	Object or finger touches the roof of the mouth → baby sucks
Tonic neck ("fencing")	Head turned to one side → arm & leg on that side extend, opposite side flex
Palmar grasp	Palm touched → baby wraps the fingers around the examiner's finger

Absent, asymmetric or **persistent primitive reflexes beyond their expected ages** need evaluation.

الفحص العصبي:

• **الترتيب**: التاريخ المرضي ← مستوى الوعي ← الأعصاب الدماغية ← الحركة ← الـ Reflexes ← الإحساس والتوازن والمشي. والفحص **الجلد** كويس، لأن الجهاز العصبي والجلد أصلهم واحد (ectoderm).

• **GCS من 3 لـ 15**: لو 12 ≤ يبقى إصابة راس شديدة، ولو 8 ≤ ممكن يحتاج حماية الـ airway. ولو الطفل مركّب أنبوبة أو لسه مبيتكلمش، الـ **Motor هو الأهم.**

• **الأعصاب الدماغية بالـ لعب**: لعبة بيتابعها بعينه (3، 4، 6 للأعصاب H pattern)، وجرس للسمع (العصب 8).

• الـ Power من 0 لـ 5، والـ Reflexes من 0 لـ 4+. الـ **Babinski طبيعي في الرضيع، وعدم التماثل أهم من الدرجة.**

5. Cardiovascular Examination

1. Look, Feel, Listen

1 · INSPECTION

— Color and distress, clubbing, precordial bulge, visible apex beat.

2 · PALPATION

— Apex beat, thrills and heaves; **radial and femoral pulses**; capillary refill; liver edge.

3 · AUSCULTATION

— S1, S2, added sounds and murmurs at all 4 areas; diaphragm and bell; sitting and lying.

AREA	SITE
Aortic	Right upper sternal border
Pulmonic	Left upper sternal border
Tricuspid	Left lower sternal border
Mitral	Apex beat

A murmur is not the whole exam: serious defects can present without one. **Always check femoral pulses and perfusion.**

2. Murmurs in Children: Innocent or Pathological?

— **< 1%** of pediatric murmurs are pathological · **50 – 80%** of children have a murmur at some point.

FAVORS INNOCENT	RED FLAGS ↠ REFER
Soft, short, systolic	Diastolic or holosystolic
Localized, varies with position	Loud murmur or thrill
Louder with fever or anemia	Abnormal pulses or cyanosis
Normal pulses, perfusion and history	Symptoms, failure to thrive · **Neonatal murmur: echo advised**

فحص القلب:

• **أماكن سماع القلب**: الـ Aortic يمين فوق، والـ Pulmonic شمال فوق، والـ Tricuspid شمال تحت، والـ Mitral عند الـ apex.

• **أقل من 1% من الـ murmurs مرضية**. البريء: خفيف وقصير وsystolic ومحدود المكان وبيتغير مع الوضع. علامات الخطر: diastolic أو holosystolic، أو عالي ومعاه thrill، أو cyanosis، أو نبض غير طبيعي. والـ murmur في حديث الولادة محتاج echo.

• **دايماً افحص الـ femoral pulses**، لأن فيه عيوب خطيرة بتيجي من غير murmur.

6. Respiratory Examination

STEP	WHAT TO DO
1. Inspection	Rate for a **full 60 seconds**, pattern, work of breathing, chest shape
2. Palpation	Chest expansion (hands on posterior chest, thumbs at midline), tracheal position, tenderness
3. Percussion	Compare sides at each level; dullness vs hyper-resonance
4. Auscultation	Breath sounds, wheeze, crackles, stridor, transmitted sounds; **compare left with right**, front and back

Signs of Respiratory Distress

INCREASED WORK OF BREATHING

— Tachypnea
— Retractions: supraclavicular, suprasternal, intercostal, subcostal
— Nasal flaring · **Head bobbing in infants** · Accessory muscle use

MODERATE TO SEVERE DISTRESS

— **Grunting** (expiratory)
— Retractions in more than one muscle group
— Cyanosis or pallor · Fatigue, lethargy, agitation

فحص الصدر:

• **التنفس يتعد دقيقة كاملة**، وقارن اليمين بالشمال في كل مستوى، من قدام ومن ورا.
• علامات زيادة المجهود: tachypnea، وretractions، وnasal flaring، و**head bobbing في الرضيع**، وaccessory muscles. والـ **Grunting** معناه distress متوسط لشديد.

7. Abdominal Examination

1. Sequence & Technique

— **1. Position & exposure:** supine, **hips flexed** to relax the wall; expose nipple to knee.
— **2. Inspection:** shape, umbilicus and hernias, distension, movement with breathing, scars.
— **3. Auscultation:** bowel sounds **before palpation**; bruits.
— **4. Percussion:** liver span, spleen, ascites.
— **5. Palpation:** light, then deep; liver, spleen, kidneys, masses; use distraction.

Palpate from the lower quadrants upward so a low liver or spleen edge is not missed. Warm hand, light touch; **watch the face**.

ORGAN	NORMAL IN CHILDREN
Liver	Edge palpable **up to 2 cm** below costal margin
Spleen	Tip may be felt **1 – 2 cm** below left costal margin (infants, young children)

2. Liver, Spleen & Ascites

TECHNIQUE

— Measure the liver edge in the **midclavicular line** with a tape.
— Define the **upper border by percussion** to exclude a pushed-down liver.
— Spleen: **bimanual palpation**, right lateral position if needed.

NEPHROLOGY PEARLS

— **Ascites:** flank dullness, shifting dullness, fluid thrill.
— Seen in **nephrotic syndrome** and chronic liver disease.
— Bimanual palpation of the flanks for **renal masses**.

فحص البطن:

• **الترتيب**: Inspection → Auscultation → Percussion → Palpation. السماعة **قبل** الجس، والطفل نايم على ضهره وورکه متني علشان البطن ترتخي.
• الجس **من تحت لفوق** علشان منفوّتش كبد أو طحال نازلين، وبُص في وش الطفل وانت بتجس.
• **الكبد طبيعي لحد 2 سم** تحت الضلوع، والطحال 1–2 سم في الرضع. والـ Ascites بالـ shifting dullness والـ fluid thrill، ويحصل في الـ nephrotic syndrome.

Supplement — Content of the Image Slides

Items shown only as images/diagrams in the lecture slides.

1. Down Syndrome — Facial Features (General Examination)

- Upslanting palpebral fissures
- Epicanthal fold
- Flattened nasal bridge
- Brushfield spots
- Low-set, small, folded ears
- Short neck

2. Tracheal Deviation

AWAY FROM THE LESION (PUSH)

- Tension pneumothorax
- Massive pleural effusion
- Mediastinal neoplasms
- Retrosternal goiter
- Upper lung tumours

TOWARDS THE LESION (PULL)

- Upper lobe collapse (atelectasis)
- Apical / upper lobe fibrosis
- Previous pneumonectomy
- Lung agenesis

3. Stages of Clubbing

STAGE	FINDING
1	Normal angle, but increased fluctuancy of the nail bed
2	Loss of the angle between nail and nail bed
3	Increased curvature of the nail
4	Expansion of the terminal phalanx (drumstick)

4. Primitive Reflexes — Age of Disappearance

REFLEX	DISAPPEARS BY	REFLEX	DISAPPEARS BY
Rooting / Sucking	~4 months	**ATNR**	~6 months
Palmar grasp	~6 months	**Plantar grasp**	~10 months
Moro	~6 months	**Babinski**	~12 months

5. Adventitious Breath Sounds

SOUND	CHARACTER	TIMING
Crackles	Popping; not cleared by coughing	Inspiration
Rhonchi	Low-pitched, coarse snoring / moaning	Mainly expiration
Wheezes	High-pitched, musical squeaking	Mainly expiration

6. Abdominal Regions & Quadrants

9 REGIONS

- Upper: R hypochondriac · Epigastric · L hypochondriac
- Middle: R lumbar · Umbilical · L lumbar
- Lower: R inguinal · Hypogastric · L inguinal

4 QUADRANTS

- RUQ — Right upper quadrant
- LUQ — Left upper quadrant
- RLQ — Right lower quadrant
- LLQ — Left lower quadrant

ملخص الصفحة:

- **Down:** عيون مايلة لفوق + epicanthal fold، أنف مفلطح، ودان صغيرة وواطية، Brushfield spots، رقبة قصيرة.
- **القصبة الهوائية:** تتزق **بعيد** عن الـ pneumothorax/effusion/الأورام، وتتشد **ناحية** الـ collapse/fibrosis/pneumonectomy.
- **Clubbing:** 4 مراحل، أولها زيادة الـ fluctuancy وآخرها الـ drumstick.
- **الـ Reflexes:** الـ Rooting يختفي بدري (4 شهور) والـ Babinski آخر واحد (12 شهر).
- **Crackles** في الشهيق ومبتروحش بالكحة، **rhonchi** واطية خشنة و**wheezes** عالية موسيقية — الاتنين أكتر في الزفير.

8. Self-Evaluation MCQs (Strictly Based on Lecture Slides)

1 **From what age should blood pressure be measured annually in children (AAP 2017)?**

- a) From birth
- **b) From 3 years ✓**
- c) From 6 years
- d) From 12 years

الإجابة: b — حسب AAP 2017 الضغط بيتقاس كل سنة من سن 3 سنين، على الدراع اليمين، والـ cuff حوالي 40% من محيط الدراع.

2 **Which growth chart is used for a child under 2 years of age?**

- **a) WHO Child Growth Standards ✓**
- b) CDC growth reference charts
- c) BMI-for-age chart
- d) No chart is needed

الإجابة: a — تحت سنتين WHO charts مع قياس الطول نايم. ومن 2 لـ 20 سنة CDC charts.

3 **A pediatric GCS of 8 or less indicates that the child:**

- a) Is fully alert
- b) Has mild head injury only
- **c) May need airway protection and ICP monitoring ✓**
- d) Should be discharged

الإجابة: c — الـ GCS من 3 لـ 15. لو 12 ≤ يبقى إصابة شديدة، ولو 8 ≤ ممكن يحتاج حماية الـ airway ومتابعة الـ ICP.

4 **Which murmur feature is a red flag requiring referral?**

- a) Soft, short systolic murmur
- b) Murmur that varies with position
- **c) Diastolic or holosystolic murmur ✓**
- d) Murmur louder with fever

الإجابة: c — أقل من 1% من الـ murmurs مرضية. الـ diastolic أو الـ holosystolic، أو العالي ومعاه thrill، دي علامات خطر.

5 **The correct sequence of the pediatric abdominal examination is:**

- a) Palpation, percussion, auscultation, inspection
- **b) Inspection, auscultation, percussion, palpation ✓**
- c) Inspection, palpation, percussion, auscultation
- d) Auscultation, inspection, palpation, percussion

الإجابة: b — الـ bowel sounds بتتسمع **قبل** الجس، والجس من الـ lower quadrants لفوق.

6 **Which sign indicates moderate to severe respiratory distress in a child?**

- a) Respiratory rate counted for 60 seconds
- **b) Expiratory grunting ✓**
- c) Normal chest expansion
- d) Calm, playful child

الإجابة: b — الـ Grunting علامة distress متوسط لشديد، ومعاه الـ retractions في أكتر من مجموعة عضلات، والـ cyanosis.

Take-Home Points

— **Observe before you touch** — much of the respiratory and general exam is visual.
— **Growth and BP are vital signs** — plot every child; measure BP from 3 years.
— **Neuro exam is play-based** — age-appropriate tests and the pediatric GCS.
— **A murmur is only one finding** — check pulses and perfusion; echo for red flags.
— **Palpate from low to high** — lower quadrants upward; measure liver and spleen.

Closing

كفارة المجلس

سُبْحَانَكَ اللَّهُمَّ وَبِحَمْدِكَ، أَشْهَدُ أَنْ لَا إِلَهَ إِلَّا أَنْتَ، أَسْتَغْفِرُكَ وَأَتُوبُ إِلَيْكَ

دعاء بعد المذاكرة

اللَّهُمَّ إِنِّي أَسْتَوْدِعُكَ مَا عَلَّمْتَنِيهِ، فَارْدُدْهُ إِلَيَّ عِنْدَ حَاجَتِي إِلَيْهِ، وَلَا تُنْسِنِيهِ يَا رَبَّ الْعَالَمِينَ

ختامًا: نسأل الله أن يجعله علمًا نافعًا، وأن يوفقك في الامتحان ويكرمك بالنتيجة التي ترضاها يا رب العالمين. آمين 🤍